Sharukshan Niranjan

Electrical and Electronic Engineering Undergraduate

Computer Vision | Remote Sensing | Multispectral Imaging

Peradeniya, Sri Lanka | +94 76 622 5553 | n.sharukshan@gmail.com | linkedin.com/in/sharuk-niran4a6b20

Research Profile

Final-year Electrical and Electronic Engineering undergraduate (GPA: 3.812/4.00) conducting research and development in computer vision, remote sensing, and multispectral imaging. Current work focuses on learning-based change detection from bi-temporal satellite imagery, with complementary experience in computational imaging for agricultural and food-quality sensing. Experienced across the research pipeline, including experimental design, data acquisition and annotation, image preprocessing, model development, reproducible evaluation, uncertainty analysis, and scientific writing. Seeking PhD opportunities in computer vision and machine learning for Earth observation and intelligent sensing.

Research Interests

Computer vision for Earth observation; remote-sensing image analysis; binary and semantic change detection; image segmentation; multispectral and computational imaging; state-space and Mamba-based vision models; robust and data-efficient learning; interpretable model evaluation.

Education

University of Peradeniya — **Feb. 2022 – Oct. 2026 (expected)**

B.Sc. Engineering (Hons.) in Electrical and Electronic Engineering — *Peradeniya, Sri Lanka*

- **Current GPA: 3.812/4.00.**
- Selected coursework: Machine Intelligence and Smart Systems, Signal Processing, Nonlinear Systems, Industrial Statistics, Embedded Systems, and Digital System Design.

G.C.E. Advanced Level Examination — **2020**

Physical Science Stream, English Medium — *Sri Lanka*

- Three A grades in Combined Mathematics, Physics, and Chemistry; Z-score: 2.0868.

Research and Professional Experience

Farbe Technologies (Pvt) Ltd — **2026 – Present**

Research and Development Intern — *Sri Lanka*

- Conduct research and development in remote-sensing change detection, with emphasis on bi-temporal satellite imagery for flood and land-cover analysis.
- Curate multi-temporal flood imagery and ground-truth masks; developed a web-based annotation workflow for efficient pixel-level labelling and dataset quality control.
- Develop and evaluate deep computer-vision pipelines, including CNN-, transformer-, and Mamba-based change-detection models, using reproducible data splits and metrics such as F1 score and intersection over union.
- Contribute to industrial computer-vision initiatives by translating operational requirements into data, experimentation, validation, and technical-documentation workflows.

University of Peradeniya — **Apr. 2025 – Present**

Undergraduate Researcher – Multispectral Imaging — *Peradeniya, Sri Lanka*

- Conduct two experimental studies on transmittance multispectral imaging for chlorophyll estimation and quantitative detection of urea adulteration in bovine milk.
- Design data-acquisition protocols and implement end-to-end pipelines for image correction, region-of-interest extraction, spectral feature engineering, regression, validation, and uncertainty analysis.
- Lead author of a peer-reviewed MERCon 2026 computer-vision paper and currently preparing a journal manuscript on MSI-based chlorophyll estimation.

Publications

Peer-reviewed conference paper

[C1] **Sharukshan Niranjan**, Iresha Ranaweera, Tharindu Chandrarathne, Kalana Dissanayaka, Roshan Godaliyadda, Vijitha Herath, Parakrama Ekanayake, and Janak Vidanarachchi, "Non-Destructive Quantification of Urea Adulteration in Bovine Milk Using Transmittance Multi-Spectral Imaging," accepted for presentation at the *12th International Moratuwa Engineering Research Conference (MERCon 2026)*, Image Processing and Computer Vision track, University of Moratuwa, Sri Lanka, Aug. 2026.

Manuscript in preparation

[J1] **Sharukshan Niranjan** et al., "Multispectral Imaging-Based Chlorophyll Estimation Using Statistical and Machine-Learning Regression," journal manuscript in preparation, 2026. *(Working title.)*

Selected Research Projects

Remote-Sensing Change Detection — **Feb. 2026 – Present**

Computer Vision, Deep Learning, Earth Observation

- Investigate binary and semantic change detection in bi-temporal satellite imagery, with current emphasis on flood-related changes and Mamba-based visual architectures.
- Build training and evaluation workflows for benchmark and curated datasets; address class imbalance, threshold selection, boundary quality, and reproducible reporting through precision, recall, specificity, F1, IoU, and confusion-matrix analysis.
- Developed an online annotation application to create and refine pixel-level change masks for supervised model development.

Non-Destructive Chlorophyll Estimation Using MSI — **Apr. 2025 – Present**

Computational Imaging, Statistical Learning

- Constructed a transmittance MSI dataset using 13 discrete bands from 365–940 nm and spectrophotometric measurements as laboratory ground truth.
- Implemented dark-current correction, ROI extraction, filtering, mean and superpixel representations, PCA/LDA analysis, spectral feature selection, KL-divergence modelling, and polynomial ridge regression.
- Achieved a validation R^2 of 0.9715 and test R^2 of 0.9805 with the selected polynomial ridge model; analysed 95% confidence/prediction intervals and propagated uncertainty to physiological estimates.
- Preparing the study as a journal manuscript, with emphasis on robust calibration and non-destructive plant-sensing applications.

Quantification of Urea Adulteration in Milk Using MSI — **Nov. 2025 – Present**

Computational Imaging, Regression

- Designed the dataset-generation protocol for 80 independently prepared image sets spanning ten density-balanced urea concentration levels and 12 spectral bands from 365–940 nm.
- Developed dark-current correction, denoising, ROI/superpixel feature extraction, and regression workflows for quantitative adulteration estimation.
- Achieved a validation R^2 of 0.9599 with linear regression and 0.9773 with a feed-forward neural network; reported the work in the MERCon 2026 paper listed above.

Technical Skills

Programming: Python, MATLAB, R, LaTeX
Machine Learning: PyTorch, scikit-learn, neural networks, regression, feature engineering, model validation
Computer Vision: OpenCV, image preprocessing, segmentation, change detection, multispectral image analysis
Scientific Computing: NumPy, pandas, SciPy, Matplotlib, PCA, LDA, statistical and uncertainty analysis
Tools: Linux, Git, Jupyter, Overleaf
Hardware: FLIR machine-vision cameras, Arduino, ATmega328P, FPGA, sensor integration, PCB design, AutoCAD

Honours and Leadership

- **Vice-President**, Ceylon University Dramatic Society, University of Peradeniya — 2024–2025
- **Second- and third-place finishes**, University of Peradeniya inter-/intra-faculty swimming tournaments 2022–2024
- **Swimming competitor**, University of Peradeniya — 2022–2024

Professional Memberships and Activities

- Student Member, Institution of Engineers, Sri Lanka (IESL) — 2025–Present

- Student Member, Electrical and Electronic Engineering Society, University of Peradeniya 2025–Present

Academic References

Prof. G.M.R.I. Godaliyadda
Professor
Department of Electrical and Electronic Engineering
University of Peradeniya, Sri Lanka
roshan@eng.pdn.ac.lk

Prof. M.P.B. Ekanayake
Professor
Department of Electrical and Electronic Engineering
University of Peradeniya, Sri Lanka
mpbe@eng.pdn.ac.lk